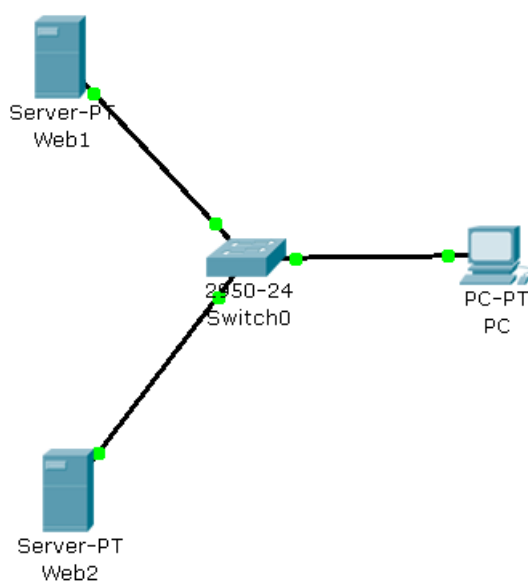




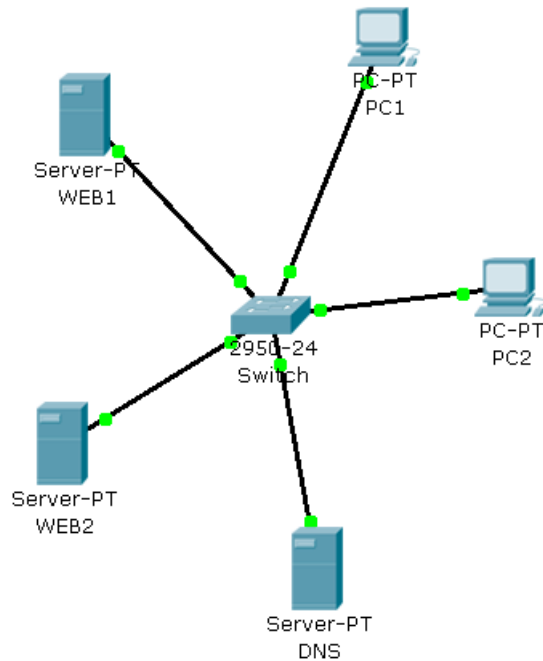
Lab 8 – HTTP & DNS

1 Part A:

During this part, we will carry out the following assemblies:



Assembly A



Assembly B

For all machines, we will use the IP addresses $10.0.0.X$ where $X \in [0;255]$ and the mask of the sub-network $255.0.0.0$.

Exercise 1 : Web Server (HTTP)

Run Packet Tracer and assembly A. Create a different web page on each server.

1. Test the arp and ping commands at the command prompt.
2. Open the web browser and try to access the web pages of the different servers.
3. Same, but this time, use the simulation mode.

Exercise 2 : DNS

Carry out assembly B under Packet Tracer by assigning to each machine an IP address.

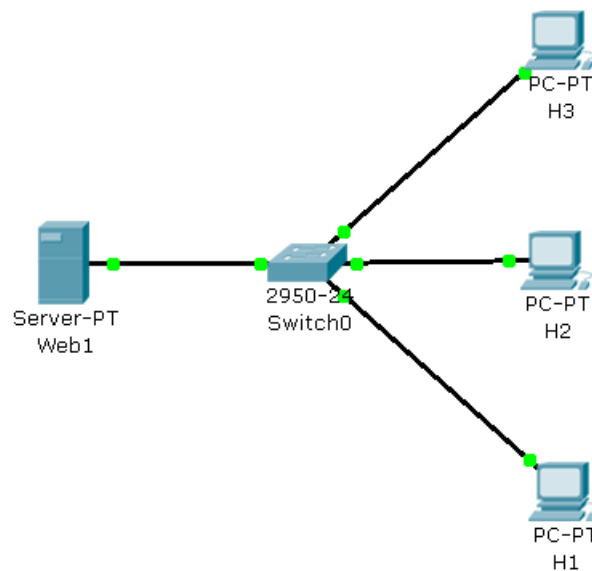
For now, ignore the DNS configuration.

1. Verify that web servers can be accessed using the ping command and a web browser.
2. We want to access web servers 1 and 2, this time using a name rather than their IP address.
 - Configure the DNS server and assign a name to each web server, as well as to each machine (name them PC1, PC2, Web1 and Web2).
 - What modifications should be made to the machine configurations?
 - Can we use the names of the machines with ping?
3. Analyze the HTTP and DNS messages generated when accessing web pages.
4. Analyze the exchanged TCP messages and note the ports used and the sequence and acknowledgment numbers.

2 Part B:

Exercise 1: Install Web server using Packet Tracer

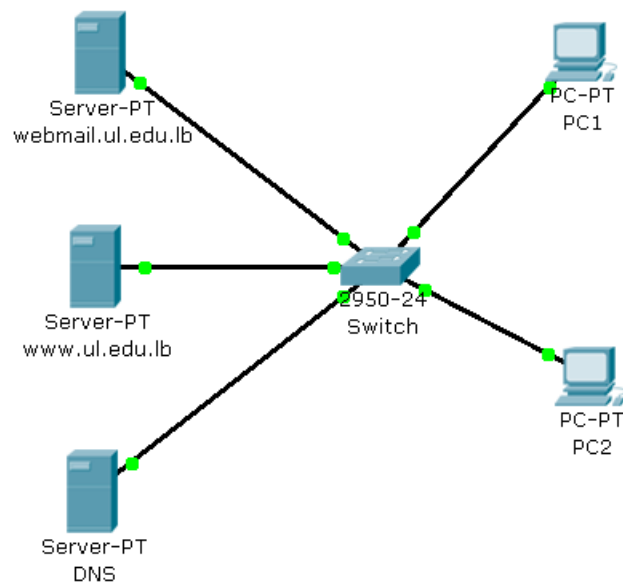
Carry out the following assembly:



1. Analyze all the messages that are exchanged when a computer accesses the Web server, using the simulation mode.
2. Configure the DNS service on the web server so that computers can access its web page by typing a URL rather than typing its IP address.
3. Separate the web server from the DNS server. What does it change?

Exercise 2: Multiple web servers

Carry out the following assembly:



You will make a simple page on the web server `www.ul.edu.lb` with a link to the web server `webmail.ul.edu.lb` and vice versa.

Annex

Some reminders on HTML

The default page for a website is named `index.html`. Here is the basic structure of a HTML file:

```
<html>
<head>
</head>
<body>
<h1>It's a title</h1>
It's a text
</body>
</html>
```

The header (between the head tags) is not required here.

Activate / configure services under Packet Tracer

In terminal equipment (End Devices category), only servers (Server-PT) can host services: HTTP, DNS, DHCP, etc. To activate and configure them, click on a server added in the work plan and go to the Config tab. To configure the HTTP service, click on the HTTP button and you will get the window shown in the figure below. From this window, you can enable / disable the HTTP service and edit the web page.

